Strategic Investment in a Forward-Looking Monetary–Fiscal Markov Game

Analytical mechanism, equilibrium construction, and parameter regions

Monetary–Fiscal Coordination Project

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Purpose of this deck

Single equilibrium concept

The dynamic model is a **feedback Markov-perfect equilibrium (MPE)**. Policy is reset each period as a function of the payoff-relevant state.

What we will do

- derive the static interaction statistic;
- show how adjustment costs create a strategic state;
- construct the recursive LQ policy game;
- embed it in a forward-looking NK private block;
- characterize parameter regions analytically.

What we will not do

- impose country-specific parameter values;
- present data-based impulse responses;
- select arbitrary numerical adjustment costs;
- infer importance from a strategic share alone.

Analytical construction



Question 1

What structural terms make one authority react strongly to the other?

Question 2

When can current policy manipulate that future reaction?

Question 3

When do private expectations magnify the feedback without destroying determinacy or stability?

A minimal static policy environment

Let the macroeconomic target vector be $y = (\pi, x)'$:

$$\boxed{y = \bar{y} + D_M m + D_F f + e.}$$

For $i \in \{M, F\}$,

$$L_i = (y - y_i^*)' W_i (y - y_i^*) + r_i a_i^2, \qquad a_M = m, \quad a_F = f.$$

Technology

D_M, D_F are signed policy-transmission vectors. Their direction determines whether the instruments reinforce or offset one another in macro space.

Preferences

W_M, W_F encode mandate differences. $r_i > 0$ is direct own-instrument curvature and guarantees a finite desired action.

Static best responses

The monetary first-order condition is

$$D_M' W_M (y - y_M^*) + r_M m = 0.$$

Substitution for y gives

$$m = \alpha_M + R_{MF} f, \qquad R_{MF} = -\frac{C_M}{H_M},$$
$$C_M = D_M' W_M D_F, \qquad H_M = r_M + D_M' W_M D_M.$$

Similarly,

$$f = \alpha_F + R_{FM} m, \qquad R_{FM} = -\frac{C_F}{H_F},$$
$$C_F = D_F' W_F D_M, \qquad H_F = r_F + D_F' W_F D_F.$$

Interpretation

C_i is cross-policy curvature under authority i's mandate; H_i is the curvature of its own decision problem. A reaction is steep when $|C_i|/H_i$ is large.

The static response product

Solving the two best responses simultaneously gives

$$\begin{bmatrix} m^N \\ f^N \end{bmatrix} = \frac{1}{1-\Gamma} \begin{bmatrix} 1 & R_{MF} \\ R_{FM} & 1 \end{bmatrix} \begin{bmatrix} \alpha_M \\ \alpha_F \end{bmatrix}, \qquad \boxed{\Gamma = R_{MF} R_{FM} = \frac{C_M C_F}{H_M H_F}.}$$

$\Gamma > 0$: reciprocal reinforcement

Reaction slopes have the same sign. As $\Gamma \to 1^-$, small desired-policy shifts produce large equilibrium actions.

$\Gamma < 0$: reciprocal dampening

Reaction slopes have opposite signs. The two responses counteract rather than compound one another.

Large mandate disagreement is insufficient unless it produces large $|C_M|$ and $|C_F|$ relative to H_M, H_F.

How shocks enter the static game

The shock e and target gaps enter α_M and α_F; the structural cross-curvatures determine the slopes.

Demand-type disturbance

Output and inflation move together.

- Authorities may want stabilization in the same direction.
- Own instrument costs create an incentive to shift the burden.
- The realized game can resemble free riding.

Cost-push disturbance

Output and inflation move in opposite directions.

- Mandate weights select different points on the trade-off.
- Each authority may offset the other's macroeconomic effect.
- The realized game can resemble an offset race.

The shock also determines whether the equilibrium state loads heavily on the direction in which strategic incentives operate.

A one-period leadership benchmark

Suppose the monetary authority chooses first and fiscal policy follows its best response:

$$f = BR_F(m) = \alpha_F + R_{FM} m.$$

The monetary leader solves

$$\min_m L_M\big(m, BR_F(m)\big)$$

and recognizes

$$\frac{dy}{dm} = D_M + D_F R_{FM}.$$

Role in the argument

This benchmark reveals the value of manipulating the rival's reaction. It does not explain where leadership comes from. The recursive MPE will endogenize the same derivative through a payoff-relevant state.

Static parameters that make manipulation valuable

The one-period leader advantage is large when:

1. the follower slope $|R_{FM}| = |C_F|/H_F$ is large;
2. the follower's action strongly affects the leader's targets, $\|D_F' W_M\|$ is large;
3. the leader can move its instrument cheaply enough, so H_M is not too large;
4. the shock/target gap makes the induced response valuable at the realized state.

Preview of the dynamic model

Replace the imposed follower response R_{FM} with the MPE derivative of future fiscal policy with respect to today's monetary action.

Why dynamics are needed

In a simultaneous policy game, neither authority can commit to a current move before the other. Leadership requires a link from current policy to future incentives.

adjustment cost:	$\lambda_i(a_{i,t} - a_{i,t-1})^2$.

Mechanical implication

$a_{i,t}$ becomes the reference point for $a_{i,t+1}$. Current policy is now a component of the next state.

Strategic implication

If the rival's feedback rule responds to that inherited position, current policy changes future rival policy.

A two-date feedback game

Let z_2 contain macro states and the date-1 instruments. At date 2,

$$m_2 = k'_{M,2} z_2, \, qquadf_2 = k'_{F,2} z_2.$$

At date 1, both authorities choose simultaneously and anticipate these state-contingent date-2 rules:

$$V_{M,1}(z_1) = \min_{m_1} \left\{ \ell_{M,1} + \beta_M V_{M,2}(z_2) \right\},$$
$$V_{F,1}(z_1) = \min_{f_1} \left\{ \ell_{F,1} + \beta_F V_{F,2}(z_2) \right\}.$$

Equilibrium concept

Backward induction produces a finite-horizon Markov-perfect equilibrium: strategies are optimal in every date-2 subgame and depend on the state inherited from date 1.

The strategic-investment derivative

Let the state evolve as

$$z_{t+1} = Az_t + B_M m_t + B_F f_t.$$

Future fiscal policy responds to today's monetary action by

$$\boxed{\frac{\partial f_{t+1}}{\partial m_t} = k_F' B_M.}$$

Decompose by state component h:

$$k_F' B_M = \sum_h k_{Fh} B_{Mh} = \underbrace{\sum_{h \in macro} k_{Fh} B_{Mh}}_{\text{macro-state channel}} + \underbrace{k_{F,m^-} B_{M,m^-}}_{\text{inherited-policy channel}}.$$

Necessary condition

Adjustment cost activates m_t as m_{t+1}^-. Strategic investment still vanishes if fiscal policy does not react to that state: $k_{F,m^-} = 0$.

The rival-response term in the MPE first-order condition

Schematically, monetary policy's continuation derivative is

$$\frac{dV_{M,t+1}}{dm_t} = \underbrace{V'_{M,z} B_M}_{\mathcal{N}_M\text{: own-state effect}} + \underbrace{\mathcal{H}^M_{f,t+1}\, k'_F B_M}_{\mathcal{S}_M\text{: rival-response effect}},$$

where $\mathcal{H}^M_{f,t+1}$ is the monetary authority's shadow loss from a marginal future fiscal action.

$$\boxed{\mathcal{S}_M = \beta_M (k'_F B_M)\, \mathcal{H}^M_{f,t+1}.}$$

Three factors

The strategic term requires state exposure B_M, a future fiscal reaction k_F, and a monetary payoff consequence $\mathcal{H}^M_f$.

Infinite-horizon recursive policy game

Let

$$z_t = (q_t, m_{t-1}, f_{t-1}, \varepsilon_t)'.$$

The Bellman equations are

$$V_M(z_t) = \min_{m_t}\{\ell_M(z_t, m_t, f_t) + \beta_M V_M(z_{t+1})\},$$
$$V_F(z_t) = \min_{f_t}\{\ell_F(z_t, m_t, f_t) + \beta_F V_F(z_{t+1})\}.$$

Conjecture

$$m_t = k_M' z_t, \; qquad f_t = k_F' z_t, \; qquad V_i(z_t) = z_t' P_i z_t.$$

Stationary MPE

Each rule is optimal given the rival's state-contingent rule, and the value matrices reproduce themselves under the induced closed-loop transition.

Coupled policy first-order conditions

For stage-loss matrices Q_M, Q_F, simultaneous policy choices solve

$$\begin{bmatrix} Q^M_{mm} + \beta_M B_M' P_M B_M & Q^M_{mf} + \beta_M B_M' P_M B_F \\ Q^F_{fm} + \beta_F B_F' P_F B_M & Q^F_{ff} + \beta_F B_F' P_F B_F \end{bmatrix} \begin{bmatrix} m_t \\ f_t \end{bmatrix} = - \begin{bmatrix} Q^M_{mz} + \beta_M B_M' P_M A \\ Q^F_{fz} + \beta_F B_F' P_F A \end{bmatrix} z_t.$$

This gives $m_t = k_M' z_t$ and $f_t = k_F' z_t$. With $A_c = A + B_M k_M' + B_F k_F'$,

$$P_i = C' Q_i C + \beta_i A_c' P_i A_c, qquad C = \begin{bmatrix} I \\ k_M' \\ k_F' \end{bmatrix}.$$

Existence requirements

The policy-response block must be nonsingular, value iterations must converge, and $\rho(A_c) < 1$ for a stable stationary solution.

Adjustment costs have two opposing effects

Define the one-period persistence share

$$p_i = \frac{\lambda_i}{D_i + \lambda_i},$$

where D_i is non-adjustment own-action curvature.

$p_i \approx 0$

The inherited instrument is almost irrelevant. There is little capacity to bind the future.

Intermediate p_i

Current policy meaningfully shifts the future reference point while retaining room to move today. Strategic investment can be large.

$p_i \approx 1$

Policy is highly durable but current movement is prohibitively costly. Commitment exists without usable investment capacity.

The expected relationship between λ_i and the strategic effect is generally **hump-shaped**.

Dynamic reciprocal response

Define the two future cross-policy derivatives

$$\varphi_{FM} = k'_F B_M, \, qquad \varphi_{MF} = k'_M B_F,$$

and their product

$$\boxed{\Omega = \varphi_{FM}\varphi_{MF}.}$$

$\Omega > 0$

The intertemporal cross-responses reinforce one another. Bilateral attempts to shape future positions can amplify.

$\Omega < 0$

The two intertemporal responses damp one another. One authority's investment induces an offsetting future response.

Ω is a useful diagnostic, not a complete stability condition: the full eigenvalues of A_c also contain macro states and own persistence.

Parameter set A: create a manipulable state

For authority i to invest strategically through its instrument, require

$$\boxed{\lambda_i > 0, qquad B_{i,a_i^-} \neq 0, qquad k_{j,a_i^-} \neq 0.}$$

1. $\lambda_i > 0$ makes the inherited action payoff relevant.
2. $B_{i,a_i^-} \neq 0$ maps today's action into tomorrow's state.
3. $k_{j,a_i^-} \neq 0$ makes the rival respond to that inherited action.

Not sufficient

These restrictions produce a strategic derivative, but the derivative can be economically negligible if the rival action barely affects authority i's loss or the relevant state is rarely activated.

Parameter set B: make the rival response steep

The cross coefficient k_{j,a_i^-} becomes large when the rival's FOC has large cross-state curvature relative to own-action curvature:

$$|k_{j,a_i^-}| \text{ large when } \frac{|\text{effect of } a_i^- \text{ on rival target gradients}|}{\text{rival own-action curvature}} \text{ is large.}$$

This is favored by:

- strong shared transmission of both instruments to output/inflation;
- mandate weights that make the inherited state important to the rival;
- low-to-moderate rival instrument penalty r_j;
- debt coefficients that transmit a_i into a state the rival stabilizes;
- sufficiently persistent states, so the response is not immediately dissipated.

Parameter set C: make the induced response valuable

Even if $|k_j' B_i|$ is large, authority i must care about the induced rival action:

$$\boxed{|\mathcal{H}^i_{j,t+1}| \text{ large.}}$$

This requires:

- the rival instrument to have a large effect on variables in W_i;
- a shock/target gap that gives those variables a large shadow value;
- mandate divergence that changes the desired policy mix, rather than merely rescaling both authorities' identical objective;
- limited direct substitutability through authority i's own instrument;
- sufficient agency patience β_i.

State dependence

LQ feedback coefficients are constant, but the realized strategic FOC wedge is linear in the state. A shock orthogonal to the relevant shadow-value direction can produce a small effect even with steep rules.

The forward-looking macro block

Let x_t be the output gap, π_t inflation, m_t monetary tightening, and f_t fiscal expansion:

$$
\begin{aligned}
x_t &= \mathbb{E}_t x_{t+1} - \sigma\big(m_t - \mathbb{E}_t \pi_{t+1}\big) + \psi f_t + d_t, \\
\pi_t &= \beta_p \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t, \\
b_{t+1} &= \rho_b b_t + \chi_m m_t + \chi_f f_t - \chi_\pi \pi_t.
\end{aligned}
$$

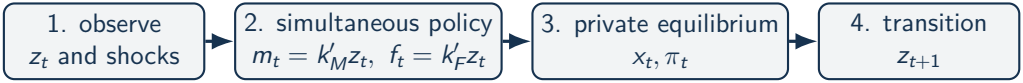
Direct policy transmission

σ, ψ, κ, and the debt coefficients determine how current instruments alter current targets and the next macro state.

Expectations transmission

Expected future output and inflation bring anticipated future policy responses into current private decisions.

Timing in the forward-looking MPE



- Private agents observe current instruments and know the stationary MPE rules that will be used in future states.
- Each authority understands current private responses and the law of motion, but cannot bind future discretion beyond the states it creates.
- No Taylor rule or fiscal rule is imposed exogenously.

Private forecasting rule

Let $y_t = (x_t, \pi_t)'$ and $q_t = (z_t', m_t, f_t)'$. Before imposing private equilibrium,

$$z_{t+1} = Nq_t + Ey_t.$$

Private agents conjecture the stationary forecast

$$\boxed{\mathbb{E}_t y_{t+1} = Gz_{t+1}.}$$

Meaning of G

G maps tomorrow's predetermined state into tomorrow's equilibrium output and inflation. Because future policy rules depend on that state, G embeds expected future monetary and fiscal behaviour.

New equilibrium loop

Policy rules depend on private transmission generated by G, while G must be validated by those same policy rules.

Reduce the private block conditional on G

Collect the IS and Phillips curves as

$$Ly_t = Dq_t + HGz_{t+1}, qquadL = \begin{bmatrix} 1 & 0 \\ -\kappa & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & \sigma \\ 0 & \beta_p \end{bmatrix}.$$

Using $z_{t+1} = Nq_t + Ey_t$,

$$\begin{aligned} M(G)y_t &= [D + HGN]q_t, \\ M(G) &= L - HGE, \\ y_t &= O(G)q_t, \qquad O(G) = M(G)^{-1}[D + HGN]. \end{aligned}$$

Within-period solvability

$M(G)$ must be nonsingular. Its smallest singular value controls how strongly beliefs about the future feed back into current private outcomes.

The policy game conditional on private beliefs

The reduced transition is

$$z_{t+1} = A(G)z_t + B_M(G)m_t + B_F(G)f_t.$$

The target equations imply belief-dependent stage-loss matrices $Q_i(G)$. Conditional on G, solve exactly the same coupled MPE system:

$$(k_M(G), k_F(G), P_M(G), P_F(G)) = \mathrm{MPE}\big(A(G), B_M(G), B_F(G), Q_M(G), Q_F(G)\big).$$

Analytical continuity

The forward-looking NK model does not replace the recursive strategic- investment game. It changes the matrices through which states and policies affect current targets and future values.

Close the rational-expectations MPE

Partition $O(G) = (O_z, O_m, O_f)$. Given the solved policy rules, the actual private outcome rule is

$$y_t = [O_z(G) + O_m(G)k_M(G)' + O_f(G)k_F(G)']\, z_t.$$

Rational expectations require

$$\boxed{G = \Phi(G) = O_z(G) + O_m(G)k_M(G)' + O_f(G)k_F(G)'.}$$

Stationary rational-expectations MPE

A solution is a fixed point in private beliefs, agency feedback rules, and agency value matrices, with a stable closed-loop state transition.

The forward-looking amplification operator

The direct sensitivity of current private outcomes to a change in expected future outcomes is governed by

$$\boxed{\mathcal{A}_E(G) = M(G)^{-1}H = [L - HGE]^{-1}H.}$$

Large amplification is associated with

$$\|\mathcal{A}_E(G)\| \text{ large} \quad \Longleftrightarrow \quad \sigma_{\min}(L - HGE) \text{ small.}$$

Magnifying parameters

High β_p, strong intertemporal demand sensitivity σ, a steep Phillips curve κ, and state-contingent future policy can strengthen the expectations loop.

Boundary

If $M(G)$ becomes singular or the full fixed point loses stability, the object is not a robust strategic equilibrium; it is a determinacy problem.

Forward-looking strategic-investment product

The inherited-policy component of authority i's strategic FOC can be written schematically as

$$\mathcal{S}_i^{policy} = \beta_i \quad \underbrace{B_{i,a_i^-}}_{\text{state creation}} \quad \underbrace{k_{j,a_i^-}}_{\text{rival response}} \quad \underbrace{\mathcal{H}_j^i}_{\text{payoff exposure}} ,$$

with $\mathcal{H}_j^i$ itself affected by

$$\underbrace{M(G)^{-1}H}_{\text{private expectations transmission}} \quad \text{and} \quad \underbrace{P_i}_{\text{discounted future shadow values}} .$$

Central result

Forward-looking behaviour is a multiplier on the economic response chain. If state creation, rival response, or payoff exposure is zero, expectations alone cannot generate strategic investment.

A useful magnitude bound

By submultiplicativity,

$$|\mathcal{S}_i| \leq \beta_i \, \|B_i\| \, \|k_j\| \, |\mathcal{H}^i_j|.$$

In the forward-looking model, a corresponding bound is

$$|\mathcal{H}^i_j| \lesssim c_{0i} + c_{1i} \left\| M(G)^{-1} H \right\| \|P_i\| \,,$$

for constants collecting mandate and transmission terms.

Four multiplicative margins

1. policy-to-state exposure $\|B_i\|$;
2. rival rule sensitivity $\|k_j\|$ in the relevant direction;
3. private expectations amplification $\left\|M^{-1}H\right\|$;
4. discounted payoff shadow value $\|P_i\|$.

Necessary restrictions for a nonzero effect

For the narrow inherited-policy channel of authority i,

$$\lambda_i > 0, \quad B_{i,a_i^-} \neq 0, \quad k_{j,a_i^-} \neq 0, \quad \mathcal{H}_j^i \neq 0, \quad z_t \neq 0 \text{ in the relevant direction.}$$

If any factor is zero:

- no adjustment cost $\Rightarrow$ no payoff-relevant inherited action;
- no state mapping $\Rightarrow$ current policy does not survive;
- no rival reaction $\Rightarrow$ only own smoothing remains;
- no payoff exposure $\Rightarrow$ the induced rival action is irrelevant;
- no state loading $\Rightarrow$ the mechanism exists in the rule but is not activated by the disturbance.

Sufficient ingredients for an economically large effect

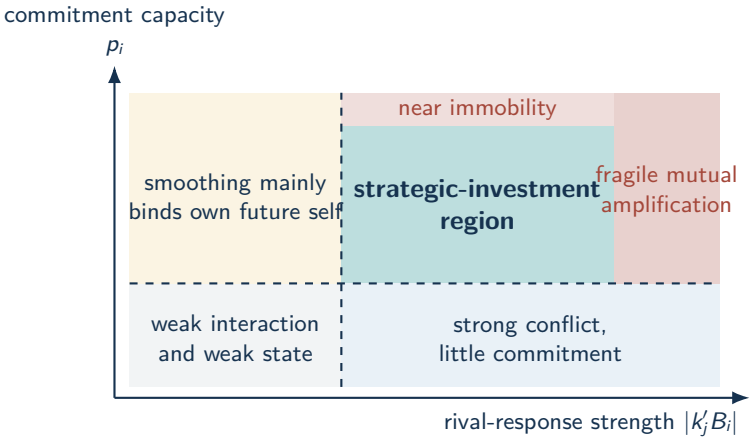
An economically large and robust effect requires the intersection of:

1. **interior commitment capacity**: p_i high enough to create persistence, but below the near-immobility region;
2. **strong cross curvature**: $|C_j|/H_j$ or its dynamic analogue is large;
3. **large payoff spillover**: the rival instrument moves targets heavily weighted by authority i;
4. **patience and durable states**: β_i and relevant persistence parameters are high;
5. **strong but determinate expectations**: $\left\|M^{-1}H\right\|$ is large while M remains nonsingular;
6. **closed-loop stability**: $\rho(A_c) < 1$ with a nontrivial margin.

Intersection, not addition

One large parameter cannot compensate for a zero elsewhere in the response chain.

Analytical phase diagram



A large robust effect lies away from both zero-persistence and near-immobility boundaries, with a strong but stable rival response.

How each parameter enters

Parameter	Direct channel	Condition for a larger strategic effect
λ_i	creates inherited-policy state	interior range: durable but still movable
r_i	own-action curvature	lower r_i steepens responses, subject to convexity/stability
σ, ψ	direct policy transmission to demand	raises cross curvature and payoff exposure if both agencies care about induced targets
κ	output-to-inflation pass-through	raises cross-target effects when inflation has shadow value
β_p	expected-inflation transmission	raises $\left\|M^{-1}H\right\|$ only if the policy-belief loop remains determinate
ρ_b, χ	durability and policy exposure of debt	useful when the rival responds to debt and debt affects the first authority's loss
β_i	policy patience	raises the weight on the future rival response
W_M, W_F	mandate curvature	must steepen cross-responses; difference alone is insufficient
shock persistence	duration of shadow-value state	raises importance when the shock loads on the strategic direction

Strategic complementarity and equilibrium multiplicity

The rational-expectations map is

$$G = \Phi(G).$$

Local uniqueness is threatened when

$$\rho\big(D\Phi(G^*)\big) \approx 1,$$

because a change in expected future outcomes induces policy and private-sector responses that almost validate the original change.

Parameters pushing toward complementarity

High private expectations transmission, persistent states, steep policy rules, low instrument curvature, and mutually reinforcing mandates.

What must still be checked

Multiple fixed points are possible, not automatic. One must establish fixed- point existence, local Jacobian properties, and closed-loop stability for each candidate equilibrium.

Do not measure strategic importance with one statistic

Use three complementary diagnostics:

$$\begin{aligned} \text{absolute wedge:} \quad & |\mathcal{S}_i|, \\ \text{bounded FOC share:} \quad & \frac{|\mathcal{S}_i|}{|\mathcal{S}_i| + |\mathcal{N}_i|}, \\ \text{equilibrium displacement:} \quad & \left\| K^{\lambda} - K^{0} \right\| \quad \text{or} \quad \left\| a_t^{\lambda} - a_t^{0} \right\|. \end{aligned}$$

Cancellation problem

A strategic share can be close to one because the nonstrategic continuation term is close to zero, even when the absolute wedge and policy displacement are tiny. Stability margins must be reported alongside all three diagnostics.

A disciplined analytical experiment

Map the model entirely in dimensionless coordinates:

1. persistence share $p_i = \lambda_i/(D_i + \lambda_i)$;
2. static response product $\Gamma = C_M C_F/(H_M H_F)$;
3. dynamic reciprocal response $\Omega = (k'_F B_M)(k'_M B_F)$;
4. expectations gain $\mathcal{E} = \left\| M(G)^{-1} H \right\|$;
5. stability margin $1 - \rho(A_c)$;
6. belief fixed-point margin $1 - \rho(D\Phi(G^*))$.

Goal

Identify regions where absolute wedges, equilibrium displacement, and stability jointly indicate meaningful strategic investment. The axes themselves are structural mechanisms, not country-specific values.

What generates a large strategic effect?

$$\text{large effect} = \text{state creation} \times \text{rival response} \times \text{payoff exposure} \times \text{expectations gain} \times \text{patience}.$$

1. Adjustment cost is necessary for the narrow inherited-policy channel, but is not itself the strategic effect.
2. The decisive object is the rival's MPE response to the state created by current policy.
3. Forward-looking private behaviour transmits that future response into current output and inflation.
4. High persistence and strong expectations can also produce immobility, instability, or indeterminacy.
5. Robust importance requires a large absolute wedge, visible equilibrium displacement, and safe stability margins.

Next analytical steps

1. Derive closed-form k_{j,a_i^-} in a reduced scalar version of the NK game.
2. Characterize signs of Γ and Ω for demand versus cost-push disturbances and alternative mandate weights.
3. Compute $D\Phi(G^*)$ to locate rational-expectations complementarity and possible multiple MPE branches.
4. Map $(p_i, \Gamma, \mathcal{E})$ jointly with both stability margins.
5. Derive boundary cases that separate negligible, large, and unstable strategic effects without assigning country-specific values.

The central task is to characterize the mechanism, its sign, and its stable parameter region.

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Questions and discussion

Backup: static Nash derivation

For authority M,

$$
\begin{aligned}
L_M &= (\bar{y} + D_M m + D_F f + e - y_M^*)' W_M (\bar{y} + D_M m + D_F f + e - y_M^*) + r_M m^2, \\
0 &= D_M' W_M (\bar{y} + D_M m + D_F f + e - y_M^*) + r_M m, \\
m &= -\frac{D_M' W_M (\bar{y} + e - y_M^*)}{H_M} - \frac{C_M}{H_M} f.
\end{aligned}
$$

Thus

$$
\alpha_M = -\frac{D_M' W_M (\bar{y} + e - y_M^*)}{H_M}, \qquad R_{MF} = -\frac{C_M}{H_M}.
$$

The fiscal expression follows symmetrically.

Backup: social welfare and coordination

Let true social loss be

$$L_S = (y - y_S^*)' W_S (y - y_S^*) + r_M^S m^2 + r_F^S f^2.$$

The cooperative static allocation solves

$$\begin{bmatrix} D_M' W_S D_M + r_M^S & D_M' W_S D_F \\ D_F' W_S D_M & D_F' W_S D_F + r_F^S \end{bmatrix} \begin{bmatrix} m \\ f \end{bmatrix} = - \begin{bmatrix} D_M' W_S \\ D_F' W_S \end{bmatrix} (\bar{y} + e - y_S^*).$$

Distinction

Coordination changes which objective is minimized. Commitment capacity changes the state transition and intertemporal feasible set. They are not substitutes.

Backup: Riccati update

Let

$$K = \begin{bmatrix} k_M' \\ k_F' \end{bmatrix}, qquad C = \begin{bmatrix} I \\ K \end{bmatrix}, qquad A_c = A + [B_M \; B_F]K.$$

For each authority,

$$P_i = C' Q_i C + \beta_i A_c' P_i A_c.$$

A stationary solution requires:

- a nonsingular simultaneous policy response block;
- finite positive-semidefinite value matrices;
- convergence of the coupled policy/value iteration;
- $\rho(A_c) < 1$.

Backup: broad versus narrow strategic channels

For monetary policy,

$$k_F' B_M = k_{F,b} B_{M,b} + k_{F,m^-} B_{M,m^-} + k_{F,f^-} B_{M,f^-} + k_{F,d} B_{M,d} + k_{F,u} B_{M,u}.$$

- $k_{F,b} B_{M,b}$: strategic response through public debt;
- $k_{F,m^-} B_{M,m^-}$: monetary inherited-policy investment;
- other terms: shock or institutional-state channels.

Interpretation

Strategic interaction can exist without adjustment costs through persistent macro states. The Petkov mechanism is the narrower inherited-policy term.

Backup: fixed-point Jacobian

At a rational-expectations equilibrium $G^* = \Phi(G^*)$, a local perturbation satisfies

$$\operatorname{vec}(\Delta G_{n+1}) \approx D\Phi(G^*)\operatorname{vec}(\Delta G_n).$$

- $\rho(D\Phi) < 1$: locally contractive belief iteration;
- $\rho(D\Phi) \approx 1$: strong expectations complementarity and slow convergence;
- $\rho(D\Phi) > 1$: the simple iteration is unstable and additional fixed points or equilibrium-selection issues become plausible.

Caution

Numerical convergence of one damped algorithm is not a global uniqueness proof.